STAT230A Lecture 4 Linear Algebra Review

1 Multivariate Normal

1.1 Definition: Standard Univariate Normal

若 random variable $Z \sim \mathcal{N}(0,1)$, 则

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

1.2 Definition: Standard Multivariate Normal

我们称 $\vec{z} = (z_1\ z_2 \ldots z_n)^\top$ 服从 standard multivariate normal (MVN), 若

$$z_1, z_2, \ldots, z_n \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$$

记作 $\vec{z} \sim \mathcal{N}_n(\vec{0}, I_n)$

1.3 Definition: General Multivariate Normal

令:

- Random vector $\vec{Y} \in \mathbb{R}^p$
- $\vec{\mu} \in \mathbb{R}^p$
- $\Sigma \in \mathbb{R}^{p \times p}$ 为 symmetric 且 positive semi-definite

则我们称 $\vec{Y}$ 服从 multivariate normal, 即 $\vec{Y} \sim \mathcal{N}_p(\vec{\mu}, \Sigma)$, 若:

$$\vec{Y} = \vec{\mu} + A\vec{Z}$$

其中,

- $\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$
- $A \in \mathbb{R}^{p \times n}$ with $AA^\top = \Sigma$ (即 $A = \Sigma^{1/2}$)

由此我们可以得到 $\vec{Y}$ 的 density:

$$f_{\vec{Y}}(\vec{y}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\vec{y}-\vec{\mu})^\top \Sigma^{-1}(\vec{y}-\vec{\mu})\right)$$

1.4 Corollary: 关于 MVN 的推论

1. 若 $\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$, 则对任意 orthogonal matrix $T \in \mathbb{R}^{n \times n}$, 有

$$T\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$$

2. 若 $\vec{Y} \sim \mathcal{N}_n(\vec{\mu}, \Sigma), B \in \mathbb{R}^{p \times n}, \vec{c} \in \mathbb{R}^p$ 则

$$B\vec{Y} + \vec{c} \sim \mathcal{N}_p(B\vec{\mu} + \vec{c}, B\Sigma B^\top)$$

3. 若:

$$\begin{bmatrix}\vec{Y}_1\\ \vec{Y}_2\end{bmatrix} \sim \mathcal{N}(\begin{bmatrix}\vec{\mu}_1\\ \vec{\mu}_2\end{bmatrix}, \begin{bmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{bmatrix})$$

则:

$$\begin{aligned}\vec{Y}_1 &\sim \mathcal{N}(\vec{\mu}_1, \Sigma_{11})\\ \vec{Y}_2 &\sim \mathcal{N}(\vec{\mu}_2, \Sigma_{22})\end{aligned}$$

且

$$\Sigma_{12} = \mathrm{Cov}(\vec{Y}_1, \vec{Y}_2) = 0 \iff \vec{Y}_1 \perp \vec{Y}_2$$

Proof: Corollary 1

由于 T 为 orthogonal matrix, 即 $T^\top T = I$, 因此有:

$$T\vec{Z} \sim \mathcal{N}_n(\vec{0}, TT^\top) \sim \mathcal{N}_n(\vec{0}, I_n)$$

Proof: Corollary 2

利用 MVN 的定义, 有:

$$\begin{aligned}B\vec{Y} + \vec{c} &= B(\vec{\mu} + \Sigma^{1/2}\vec{Z}) + \vec{c}\\ &= (B\vec{\mu} + \vec{c}) + (B\Sigma^{1/2})\vec{Z}\\ &\sim \mathcal{N}_p(B\vec{\mu} + \vec{c}, B\Sigma B^\top)\end{aligned}$$

2 Quadratic Form of a Random Vector

2.1 Definition: Quadratic Form of a Random Vector

令:

- $\vec{Y} \in \mathbb{R}^p$ 为 random vector
- $A \in \mathbb{R}^{p \times p}$ 为 square and symmetric constant matrix
  则 $\vec{Y}$ 的 quadratic form 被定义为:

$$\vec{Y}^\top A \vec{Y} \in \mathbb{R}$$

2.2 Theorem: Quadratic Form 的 Expectation

令:

- $\vec{Y} \in \mathbb{R}^p$
- $\mathbb{E}[\vec{Y}] = \vec{\mu}$
- $\mathrm{Cov}(\vec{Y}) = \Sigma$
  则:

$$\mathbb{E}[\vec{Y}^\top A \vec{Y}] = \vec{\mu}^\top A \vec{\mu} + \mathrm{Tr}(A\Sigma)$$

Proof

根据 covariance matrix 的定义, 可以得到

$$\mathbb{E}[\vec{Y}\vec{Y}^\top] = \mathrm{Cov}(\vec{Y}) + \vec{\mu}\vec{\mu}^\top = \Sigma + \vec{\mu}\vec{\mu}^\top$$

根据 linearity of expectation, 若 W 为任一 square random matrix, 则有

$$\mathbb{E}[\mathrm{Tr}(W)] = \mathrm{Tr}(\mathbb{E}[W])$$

综上, 可以得到

$$\begin{aligned}
\mathbb{E}[\vec{Y}^\top A \vec{Y}] &= \mathbb{E}[\mathrm{Tr}(\vec{Y}^\top A \vec{Y})] \\
&= \mathbb{E}[\mathrm{Tr}(A\vec{Y}\vec{Y}^\top)] \\
&= \mathrm{Tr}(A\mathbb{E}[\vec{Y}\vec{Y}^\top]) \\
&= \mathrm{Tr}(A(\Sigma + \vec{\mu}\vec{\mu}^\top)) \\
&= \mathrm{Tr}(A\Sigma) + \mathrm{Tr}(A\vec{\mu}\vec{\mu}^\top) \\
&= \mathrm{Tr}(A\Sigma) + \mathrm{Tr}(\vec{\mu}^\top A \vec{\mu}) \\
&= \mathrm{Tr}(A\Sigma) + \vec{\mu}^\top A \vec{\mu}
\end{aligned}$$

Remark: General Random Vector 的 Quadratic Form 的 Variance

对于 general random vector $\vec{Y}$, 其 quadratic form $\vec{Y}^\top A \vec{Y}$ 的 variance 较难求出, 但若 $\vec{Y}$ 为 MVN, 则可以求出

2.3 Definition: χ^2 distribution

我们称 random variable $w \in \mathbb{R}$ 服从 χ^2_k distribution, 若

$$w = \sum_{i=1}^{n} z_i^2$$

其中 $z_i \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$

Remark: 等价定义

另一种等价定义为:

$$w = \vec{Z}^\top \vec{Z}$$

其中 $\vec{Z} \sim \mathcal{N}_k(0, I_k)$

Remark: χ^2 distribution 的 mean 和 variance

- $\mathbb{E}[\chi_p^2] = p$
- $\mathrm{Var}[\chi_p^2] = 2p$

2.4 Theorem: 关于 χ^2 distribution 的定理

1. 若 $\vec{Y} \sim \mathcal{N}_n(\vec{\mu}, \Sigma)$ 且 $\Sigma \succ 0$, 则

$$(\vec{Y} - \vec{\mu})^\top \Sigma^{-1} (\vec{Y} - \vec{\mu}) \sim \chi_n^2$$

2. 若 $\vec{Y} \sim \mathcal{N}_n(0, I_n)$ 且 $H \in \mathbb{R}^{n \times n}$ 为 projection matrix with $\mathrm{rank}(H) = k \leq n$, 则

$$\vec{Y}^\top H \vec{Y} \sim \chi_k^2$$

3. 若 $\vec{Y} \sim \mathcal{N}_n(\vec{0}, H)$, 其中 $H \in \mathbb{R}^{n \times n}$ 为 projection matrix with $\mathrm{rank}(H) = k \leq n$, 则

$$\vec{Y}^\top \vec{Y} \sim \chi_k^2$$

Proof: Theorem 1

由于 $\Sigma \succ 0$, 则存在 invertible matrix $\Sigma^{1/2}$, 使得:

$$\Sigma^{1/2} (\Sigma^{1/2})^\top = \Sigma$$

因此有:

$$\Sigma^{-1} = \left((\Sigma^{1/2})^\top\right)^{-1} (\Sigma^{1/2})^{-1}$$

根据 MVN 的定义, 有:

$$\vec{Y} = \vec{\mu} + \Sigma^{1/2} \vec{Z}$$

其中 $\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$ 为 standard normal random vector

因此:

$$\begin{aligned}
(\vec{Y} - \vec{\mu})^\top \Sigma^{-1} (\vec{Y} - \vec{\mu}) &= \vec{Z}^\top (\Sigma^{1/2})^\top \Sigma^{-1} \Sigma^{1/2} \vec{Z} \\
&= \vec{Z}^\top (\Sigma^{1/2})^\top \left((\Sigma^{1/2})^\top\right)^{-1} (\Sigma^{1/2})^{-1} \Sigma^{1/2} \vec{Z} \\
&= \vec{Z}^\top \vec{Z} \\
&\sim \chi_n^2
\end{aligned}$$

Proof: Theorem 2

由于 H 为 projection matrix, 因此 H 为 symmetric matrix, 因此有以下 eigendecomposition:

$$H = P^\top \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix} P$$

其中对角阵中有 k 个 1, $n-k$ 个 0

因此有

$$\vec{Y}^\top H \vec{Y}^\top = \vec{Y}^\top P^\top \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix} P\vec{Y} = \vec{Z}^\top \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix} \vec{Z} = \sum_{i=1}^{k} z_i^2 \sim \chi_k^2$$

其中

- $\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$ 为 standard normal random vector
- $z_i \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$ 为 standard normal random variables

Proof: Theorem 3

根据 MVN 的定义, 有:

$$\vec{Y} = H^{1/2}\vec{Z}$$

其中 $\vec{Z} \sim \mathcal{N}_n(\vec{0}, I_n)$ 为 standard normal random vector

由于 H 为 projection matrix, 因此 H 所有的 eigenvalues 为 0 或 1, 有以下 eigendecomposition:

$$H = U \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} U^\top$$

其中 U 为 orthogonal matrix

因此有:

$$H^{1/2} = U \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}^{1/2} U^\top = U \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} U^\top = H$$

因此有:

$$\begin{aligned} \vec{Y}^\top \vec{Y} &= \vec{Z}^\top (H^{1/2})^\top H^{1/2} \vec{Z} \\ &= \vec{Z}^\top H^\top H \vec{Z} \\ &= \vec{Z}^\top H \vec{Z} \quad (H \text{ idempotent} \Rightarrow H^\top = H, H^2 = H) \\ &= \vec{Z}^\top U \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} U^\top \vec{Z} \\ &= \vec{Z}' \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \vec{Z}' \\ &= \sum_{i=1}^{k} (Z_i')^2 \quad (Z_i' \overset{i.i.d.}{\sim} \mathcal{N}(0,1)) \\ &\sim \chi_k^2 \end{aligned}$$